

The full employment output is given by the production capacity of the economy. It is derived from the production function (see Unit 1). Suppose the full employment output (potential output) of the economy is given by Y^* . We depict it by a vertical straight line in Fig. 3.7. The IS_0 curve and LM_0 curve intersect at E which corresponds to the full employment output.

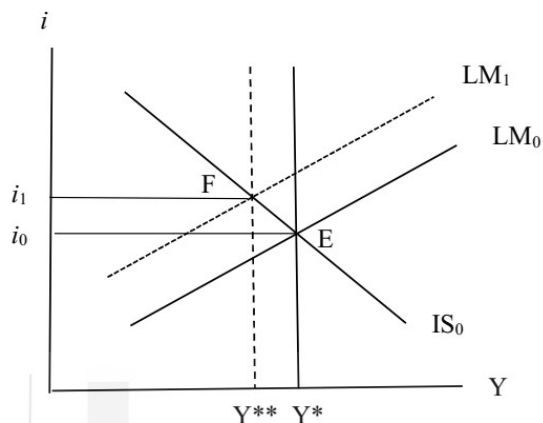


Fig. 3.7: Adjustment to Adverse Supply Shocks

Suppose the economy gets an adverse supply shock (for example, decline in agricultural productivity due to drought), as a result of which there is a decrease in the potential output of the economy from Y^* to Y^{**} . As pointed out above, the IS curve is relatively stable in such situations and the economy adjusts through a shift in the LM curve. Due to the adverse supply shock, there is a rise in prices and consequent leftward shift in the LM curve. Thus, there is a rise in the rate of interest and the new equilibrium is at F.

Check Your Progress 2

- 1) Explain how the LM curve is derived.

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- 2) Suppose the LM curve is vertical. What are its implications?

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- 3) In the IS-LM model, explain how adjustment to supply shocks takes place.

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3.5 AD-AS MODEL

In the IS-LM model we assumed that price level is constant. Let us relax that assumption and look into the impact of prices on equilibrium output in the economy. First, we derive the aggregate demand (AD) curve. Subsequently, we juxtapose the AD curve with the aggregate supply (AS) curve and find out the equilibrium level of output.

3.5.1 Derivation of the AD Curve

A change in the price level changes the real money supply ($\frac{M^s}{P}$) in the economy. This will result in a shift in the LM curve if money supply (M) is assumed to be constant. Thus, there will be a change in the equilibrium rate of interest and output. If we assume that the price level keeps changing over time, we will obtain a series of equilibrium levels of output corresponding to different price levels. If we plot such combinations of prices and output, we obtain the aggregate demand (AD) curve.

The derivation of the AD curve from IS-LM curves is shown in Fig. 3.8. In panel (a) of Fig. 3.8 we juxtapose IS_0 and LM_0 curves intersecting at point E with an equilibrium income level of Y_0 and interest rate i_0 . The LM_0 curve is based on real money stock of M_0/P_0 . Thus, we describe the combination P_0 and Y_0 as point 'a' on the AD curve in panel (b) of Fig. 3.8.

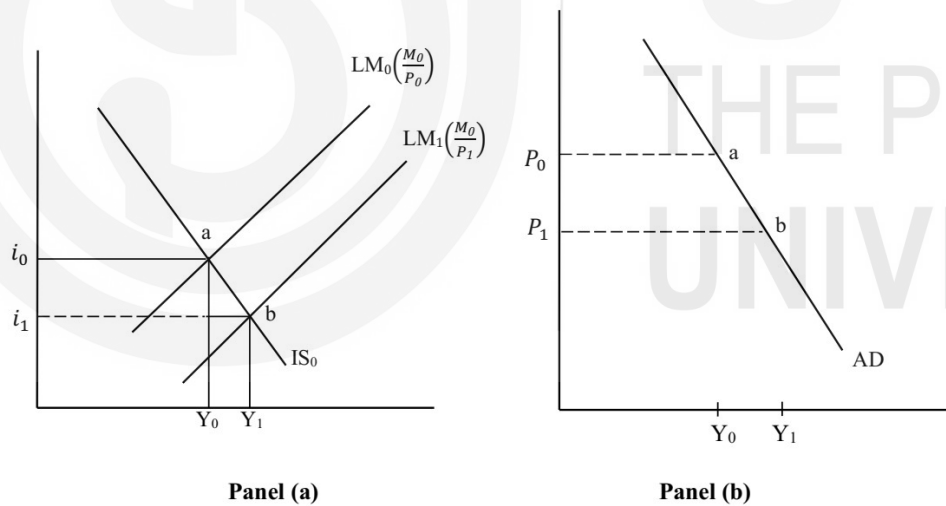


Fig. 3.8: Derivation of AD Curve

Now, let us assume that price level declines to P_1 . If there is no change in money stock, supply of real balances increases to M_0/P_1 . Note that the increase in real balances is due to decrease in prices, not due to an increase nominal money stock. The LM curve shifts to the right. In panel (a) of Fig. 3.8 we depict the new LM curve as LM_1 . With IS_0 and LM_1 , the new equilibrium point is 'b'; the rate of interest falls to i_1 and income rises to Y_1 . The combination of P_1 and Y_1 becomes point 'b' in Panel (b) of Fig. 3.7. By joining all such points as 'a' and 'b', we obtain the AD curve.

The AD curve is downward-sloping, which indicates an inverse relationship between prices and output. A decline in price level increases the real money supply, which creates disequilibrium in the money market. Increase in the supply of real balances lead to a decrease in interest rate. A decrease in the rate of interest restores equilibrium in the money market. A decrease in the rate of interest leads to an increase in the level of investment. An increase in the level of investment leads to an increase in aggregate expenditure, thereby increasing the level of output.

The AD curve is flatter, if a given change in price leads to a larger change in output. On the other hand, the AD curve is steeper if a given change in price leads to a smaller increase in output. The AD curve shifts upward to the right as a result of expansionary fiscal policy (e.g., increase in government expenditure, reduction in tax rate). A contractionary fiscal policy, on the other hand, shifts the AD curve downward to the left. Similarly, an expansionary monetary policy (e.g., reduction in interest rate, increase in money supply through open market operations, reserve requirements, etc.) will shift the AD curve to the right while a contractionary monetary policy will shift the AD curve to the left.

You should note that any factor that leads to a shift in the IS curve or the LM curve to the right will shift the AD curve to the right side. Similarly, any factor that shifts the IS curve or LM curve to the left will shift the AD curve to the left. Thus, an increase in any component of autonomous spending such as an increase in government spending leads to a rightward shift of the AD curve. Also, an increase in the nominal money supply will shift the AD curve upwards.

3.5.2 Aggregate Supply (AS) Curve

From microeconomics we know that the AS curve is upward-sloping – when price of a product increases, firms produce more of the product. In the case of the whole economy, production capacity is limited. Resources available to the economy are not unlimited. The behaviour of the AS curve is different in the long run compared to the short run. Short run, as you know, is a time period in which wages and certain other economic variables do not respond to changes in economic environment. In the long run is a time period in which wages and prices are flexible.

Classical theory discussed in Unit 1 assumed that the AS curve is vertical at the full employment level. Thus, output is perfectly inelastic to changes in prices. Keynes went to the other extreme and assumed that the AS curve is horizontal. An implication of the horizontal AS curve is that firms can increase production to any level without affecting prices. Both the versions (vertical and horizontal) of the AS curve seems unrealistic. As you will learn in later units, the new-classical economists suggest that the AS curve is upward-sloping in the short run. Lucas' in his 'aggregate supply hypothesis' advocate that economic agents respond to price and wage incentives, which influences their trade-off between leisure and work. As a result, when actual wage is higher than equilibrium wage rate, households supply more labour. Conversely, when actual wage rate is lower than equilibrium wage rate, household will decrease their supply of labour. As a result of this, the